

America on Record

A History of Recorded Sound

Second Edition



Andre Millard

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With Thomas Edison's invention of the phonograph, the beautiful music that was the preserve of the wealthy became a mass-produced consumer good – a sound recording – cheap enough to be available to all. In 1877 Edison dreamed that one day there would be a talking machine in every home, but even his legendary vision could not have foreseen the way that recorded sound would pervade modern life.

America on Record: A History of Recorded Sound provides a history of sound recording from the first thin sheet of tinfoil that was manipulated into retaining sound to the home recordings of rappers in the 1980s and the high-tech studios of the 1990s. This book examines the important technical developments of acoustic, electric, and digital sound reproduction while outlining the cultural impact of recorded music and movies. This second edition brings the story up to date, describing the digital revolution of sound recording with the rise of computers, Napster, DVDs, MP3, and iPods.

Andre Millard is Director of American Studies and Professor of History at the University of Alabama at Birmingham. He is the author of *Edison and the Business of Innovation* and a contributor to National Public Radio's *Lost and Found Sound*.

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ANDRE MILLARD

University of Alabama at Birmingham



CAMBRIDGE
UNIVERSITY PRESS

CAMBRIDGE UNIVERSITY PRESS

Cambridge, New York, Melbourne, Madrid, Cape Town,
Singapore, São Paulo, Delhi, Tokyo, Mexico City

Cambridge University Press
The Edinburgh Building, Cambridge CB2 8RU, UK

Published in the United States of America by
Cambridge University Press, New York

www.cambridge.org

Information on this title: www.cambridge.org/9780521835152

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First published 1995

Second edition published 2005

A catalogue record for this publication is available from the British Library

Library of Congress cataloguing in publication data

Millard, A. J.

America on record : a history of recorded sound / Andre Millard.— 2nd ed.
p. cm.

Includes discography (p.), bibliographical references (p.), and index.

ISBN 0-521-83515-1 (hardcover) – ISBN 0-521-54281-2 (pbk.)

1. Phonograph. 2. Sound recordings – United States.

3. Music – United States – History and criticism. I. Title.

ML1055.M47 2005

621.389'3'0973 – dc22

2005025304

ISBN 978-0-521-83515-2 Hardback

ISBN 978-0-521-54281-4 Paperback

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For my mother, Stella Mary Millard, 1923–2004

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Preface to the second edition

In the ten years since the publication of this book there have been momentous changes in the technology of recorded sound. Changes so rapid and so far reaching that some commentators have already anticipated the end of hard media like CDs and the major record companies that produce them. This edition seeks to bring the reader up to date with the new technology, new businesses, and new status of sound recordings in the digital age. In the Preface to the first edition I hoped that technological change would not make some of the machines I described obsolete before the book reached its readers. I do so again, confident that the Internet, MP3, and personal computers will not go the way of DCC or DAT. In this edition I have tried to look a bit farther into the future. These predictions are based on the beliefs that copying of digital content will not be eliminated by legal or technological means, that proprietary interests will bow to the imperative of compatibility, and that large corporations will continue to adapt to technological change. If these conditions hold I expect that this edition will last at least ten more years.

I want to thank Steve Klein and Erik Lizee for reading the manuscript and making useful suggestions and UAB graduate students Thomas Scales and Nilanjana Majumdar for helping in the production of the text.

Preface to the first edition

America on Record is an attempt to describe the technological development and cultural impact of sound recording from its inception to the early 1990s – “from Edison’s phonograph to gangsta rap,” as one reviewer put it. As space is limited, I have not provided a full account of the technology or its cultural effects – I leave this to the experts – but instead a concise narrative which covers the significant events and explains the relationship of these two important forces. This is an interdisciplinary book intended for students in History and American Studies, and their pressing schedules have been uppermost in my mind during the preparation of the manuscript.

The history was written during a time of rapid technological advance, and I have no doubt that more innovations in digital sound recording will have been made before the book appears in print. I ask the reader’s indulgence for sections that may appear outdated and hope that the machines described in the closing chapters have not become obsolete in the short time from manuscript to published book.

Researching and writing this book was truly a pleasure. I owe a great debt to Raymond Wyman, Professor Emeritus of the University of Massachusetts at Amherst, who not only read the manuscript several times but also gave me the benefit of his years of experience with audio visual technology. Peter Copeland, recording engineer and Conservation Manager of the British Library National Sound Archive in London, also went through the manuscript and helped me understand much of the technology. My friend Ed Pershey of the Tsongas Industrial History Center was full of his usual insights and guided me along the path to a readable book.

Several important archives were made available to me. I want to thank the staff of the Edison National Historic Site, especially Doug Tarr and Jerry Fabris; Sam Brylawski of the Recorded Sound Reference Center at the Library of Congress; Dan Morgenstern of the

Institute of Jazz Studies, Newark; Emmett Chisum of the American Heritage Center at the University of Wyoming, Laramie; Dick Cooper of the Alabama Music Hall of Fame; and Tina McCarthy of the Sony/CBS music archives, New York. Ruth Edge and her staff at the EMI archives in Hayes near London were most helpful, and I still have fond memories of the EMI canteen. Chris Moulton of the Listening Service of the British Library patiently provided me with hours of taped oral histories. Sheldon Hochheiser of the AT&T archives in New Jersey not only guided me through that excellent collection but also reviewed the sections on the transition to sound in motion pictures and gave me some useful advice.

Many of my colleagues were kind enough to review parts of the manuscript. My friends Paul Israel, Greg Field (both of the Edison Papers), Mark Clark, David Morton, Larry Wharton, and Virginia Whatley Smith willingly gave me their time and expertise. Allen Koenigsberg looked at the early sections on the talking machine and cleared up many of the errors. Roger Thorne and Domenic Coombe spent many pleasant hours with me talking about early recordings and playing them for me. Joanne Nathan of Unique Recording Studios was very generous with her time and assistance, as were Kim Adamian of Sony Electronics and Mary Ison of the Prints and Photographs Division at the Library of Congress.

Kevin Gustin did the computer work on the composite picture of the record catalogues (Figure 4.4), and Richard Anderson drew the diagrams. I thank them both for their fine work.

The staff of Sterne Library, University of Alabama at Birmingham, especially Gordon Dunkin, Marilyn Grush, Eddie Lusker, and Bonnie Ledbetter, gave me their usual friendly, expert assistance. A final word of thanks to Debbie Givens, center of gravity and administrator of the History Department at UAB.

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Introduction

Recorded sound is surely one of the great conveniences of modern life. We can conjure up sounds at will from a talking machine, not just the sound of our own voices but the finest music ever made, and all with the convenience of a touch of a button. In a world oppressed by the consequences of progress, the phonograph and its descendants have provided us with cheap and plentiful distraction in the comfort of our own homes. It has made living in small, windowless, air-conditioned rooms a little easier, replacing the shared Victorian pleasures of bandstand and music hall with the solitary delight of a private world of sound.

The novelty of hearing a recording of one's voice is a little over 100 years old. In November 1877 Thomas Edison invented the phonograph in his Menlo Park laboratory, a feat which earned him worldwide fame and the nickname of "The Wizard of Menlo Park." Although the term *phonograph* had been coined years before, this famous invention marked the real beginning of recorded sound technology. To the millions of Americans and Europeans who marvelled at what they thought was a great scientific achievement, Edison's phonograph perfectly represented the new machines which were changing their lives: the telephone, electric light, safety razor, street car, camera, and automobile. The phonograph provided a service which had been unimaginable to its listeners before 1877 – it was truly a modern technology.

In the late nineteenth century, it was not only inventors who looked into the future and saw a marvelous new age created by machines. In 1888 the writer and social reformer Edward Bellamy published *Looking Backward, 2000–1887*, a view of a future society "so simple and logical" in which many of the burdens of industrial America had been lifted. One of the features of this utopia was that the finest music was piped into homes by telephone wires from special studios where professional musicians played around the clock. Bellamy anticipated re-

corded sound when he described “An arrangement for providing everybody with music in their homes, perfect in quality, unlimited in quantity, suited to every mood, and beginning and ceasing at will.”¹

Thomas Edison was also a visionary, a man who made it his business to see into the future, and who often made reckless claims about the potential of his inventions well before he had even made them work. Flushed with the pride of inventing the phonograph, he made the most astounding claims for it, which anticipated Bellamy’s prophecies. Edison thought it would become commonplace, one of life’s universal pleasures. The beautiful music which was the preserve of the very rich, available to only a few lucky city dwellers, was to become a mass-produced consumer good – a sound recording – which would be cheap enough to be available to all. The talking machine would be as useful as the telephone.

From the vantage point of the late twentieth century, all these wild predictions now seem obvious: the talking machine has become an essential part of life in an industrial society. In 1877 Edison dreamed that one day in the distant future there would be a talking machine in every home. A hundred years later, the average house would contain two or three. In 1977 there was hardly a living room or den in the United States and Western Europe which did not have a home “stereo” made up of amplifier, record player, radio, and tape recorder. Most homes also had a smaller system, a portable record or tape player, in the childrens’ rooms or in the kitchen.

By the 1970s the universal medium of recorded sound was the compact tape cassette. No less an engineering wonder than the microgroove long-playing disc it superseded, the cassette had the advantages of small size and easy handling. The recorded sound of a cassette tape emerges from countless transistorized machines, from the wake-up call in the bedside alarm clock to the telephone answering machine. It can be found in numerous child’s toys, in the dashboard of the car in the garage, and even in a waterproofed unit in the shower. The tape cassette has replaced the revolving disc as the most widely used format for recorded sound. It can be heard all over the globe. In the developing nations of Africa and Asia, it is an eagerly adopted Western technology. In advanced industrial societies, there are more cassette players than there are households.²

Recorded sound has expanded well beyond the phonograph invented by Edison. It is a major source of the programs heard on the

radio, for the tradition of live broadcasting declined after World War II. The same is true for television, which abandoned live programs very quickly. Most of its programs are recorded, and the video tape which saves the image also saves the sound, including the commercial messages and the “canned” laughter, which is an indispensable part of television entertainment.

It is easy to forget that recorded sound played a major role in motion pictures, another modern technology that began in Thomas Edison’s fertile laboratory. He followed the invention of the phonograph with that of the motion picture camera; his idea was for recorded sound and images to go together. They still do. Sound is an integral part of the movie experience, and film sound has often been at the leading edge of recorded sound technology. The latest sound-reproducing systems are introduced to the public in film theaters. Stereophonic sound was developed for use in theaters in the 1930s. Quadrophonic sound was first experienced at the movies. The technologies developed to make movies talk have been successfully applied to the recording of popular music. Our home stereos are scaled-down versions of the mighty sound systems found in film theaters.

Recorded sound is not confined to the home or places of entertainment. The drive to work is usually not done in silence because the car radio, cassette, or compact disc player comes as standard equipment in the American automobile. Recorded sound also soothes the brow of the office worker as he or she takes the elevator to one of the numerous floors of the skyscraper – the typical work place of the twentieth century. Working in America is often done to the accompaniment of recorded music, which is very carefully chosen to ensure the productivity of the listener.

The widespread broadcast of recorded sound in the work place began in World War II, when it was discovered that music decreased the boredom associated with continuous, repetitive assembly-line work. The British government commissioned an important study on “Fatigue and Boredom in Repetitive Work,” which found that the man watching the bottle-capping machine or the woman soldering connections became tired not only from physical exertion but because of boredom. Music distracted the workers from their tasks without taking their eyes off the job.

Employers who played music in factories found that instead of leav-

ing their benches or machines, workers were more apt to be found whistling or singing while they worked. The management took special care to fit the tempo of the music to the pace of the work. Martial and “peppy” music was used to get employees ready for work, while slow, sentimental ballads were never played on rainy days. The main qualification was for music to be “easily recognized, easily sung, easily heard.”³ This was the beginning of the insidious “easy listening” music which awaits us in elevators and restaurants and fills in the time while we wait to be connected to telephone lines.

Background music came of age during World War II. By decreasing fatigue and raising morale, it made significant contributions to productivity, and some estimates found it increased output by as much as 25 percent.⁴ This revelation was to have enormous impact on the history of recorded sound. In the short run it brought music into the factory and office; in the long run it completely altered the sonic environment in America, leading to the omnipresence of canned music in every aspect of life. In the past, manual labor in the United States had been eased by work songs, the chants of the slaves, field hollers, or whistling while you worked in a textile mill. In the twentieth century the anonymous sounds of the talking machine usurped this function.

In both public and private places, the tyranny of recorded sound is hard to escape. Eating, shopping, and travelling are discretely serenaded by endless loops of magnetic tape. It is difficult to think of a place that does not have a concealed loudspeaker issuing forth the soothing sounds of recorded music. Even churches have succumbed to using tape-recorded bells in place of the real thing.

Edison and the other inventors of recorded sound were so enthusiastic about their achievements because the real thing in the 1870s was restricted to the great concert halls in a few major cities and music rooms in the homes of the very rich. Although the technology of recorded sound was international in nature – it came about as the result of the transatlantic movement of men and ideas – its development occurred in the United States. What especially pleased Edison was that it democratized entertainment. It was fitting that Yankee ingenuity should provide the means to put opera into the parlors of working men (as Edison always liked to claim) and make good music an affordable consumer good.

Yet Edison could not have predicted, or even imagined, that re-

corded sound would be so omnipresent in twentieth-century industrial society. Recorded music is a constant in the home, the work place (whether in office or factory), and in public places. It is heard in supermarkets, malls, trains and planes. This constant background noise, often called *Muzak* after the company that pioneered recorded music in the work place, has led some unwilling listeners to question the benefits of recorded sound. It was “synthetic music,” as Aldous Huxley called it in *Brave New World*, which could just as easily pacify as enlighten the masses. What had started as a machine to bring beautiful music to silent places, to bring the art of the concert hall into the living room, became corrupted into a device which saturated the senses with an unrelenting bombardment of recorded sounds. And Edison, who happily retreated behind the wall of silence brought by his deafness, would probably have been one of the first to denounce it.

Business and technology

The growth of the technology of recorded sound and the business enterprise based upon it exceeded even Edison’s grandiose plans for the phonograph. Putting a talking machine into every home and supplying it with records became a big business in America. One hundred years after the thin sheet of tin foil was teased and tortured into retaining sounds of speech, a vast international industry produced players and recordings by the millions.

In the United States alone, about \$3 billion worth of recordings were sold in 1977 (at retail prices). These were played on an estimated 75 million record players (playback units) in American homes. In 1976 over 3 million portable and table phonographs were sold by American manufacturers, and over 2 million phonographs and 17 million tape recorders were imported into the United States.⁵ As the centennial of Edison’s invention approached, Americans purchased 273 million long-playing records, 190 million 45-rpm singles, and over 127 million cassette and eight-track tapes.

This book is an account of the growth of this industry, a history which by necessity reflects the technological development of recorded sound, from the tin foil wrapped around Edison’s phonograph to the compact disc. This is a story primarily of change, for the industry built on the phonograph was driven forward by the constant disrupt-

tion of innovation: new materials for records, new systems of recording, new kinds of machines, and new types of recorded music. In an intensely competitive business, one invention after another arrived to upset the fragile balance between the great companies and change the relationships of the old powers with the new.

The industry of recorded sound reflects the broad themes of American business history. The small, independent companies founded by the inventors evolved into big businesses at the turn of the century, when three large companies dominated the industry. These organizations became even larger during the 1920s and 1930s when dramatic advances in recording technology made it easier to establish entertainment empires. Some of the most powerful and innovative American companies were players in this drama, such as AT&T, General Electric, RCA, and Warner Bros. Their mastery of the technology of recording made it possible for them to dominate several branches of the immensely profitable business of popular entertainment.

American industrial dominance came to an end in the postwar decades, and Japanese companies proved to be fierce competitors in the consumer electronics market. Companies like Sony and Matsushita were responsible for many critical innovations. Big Business grew ever bigger in the 1970s and 1980s as a wave of mergers produced vast, integrated media empires. Multinational organizations such as Sony and Philips then dominated a global business in which recorded-sound products were but one element in the mechanization of entertainment.

The technological history of the phonograph breaks down into three parts, each reflecting a different method of recording sound. The acoustic era began in 1877 with Edison's invention of the phonograph and ended in the late 1920s when a new system of electrical recording superseded it. The electrical era of the 1930s and 1940s was represented by the 78-rpm shellac disc and the vacuum-tube phonograph or radio/phonograph combination. Its technological high point was reached in the 1950s and 1960s with microgroove vinyl discs (the 45-rpm single and the 33-rpm long player) and the record player (based on transistors instead of vacuum tubes). By the 1970s magnetic tape slowly overtook the revolving disc as the main form of recording sound. The 1977 sales year reflected the rapid growth of sales of prerecorded cassette tapes and pointed to the dominance of this for-

mat in the years that followed. The digital era began in 1982 with the commercial introduction of the compact disc (CD).

The technological history of recorded sound also covers the history of motion pictures. The electrical era began with the first sound pictures. The techniques and equipment perfected in film studios for use in the "talkies" slowly spread to recording studios. In the 1950s magnetic tape was used to save television pictures as well as sound. The first digital recording equipment of the 1970s was based on video recorders for television.

The development of recorded-sound technology was often the result of the diffusion of ideas and techniques between film makers and record companies. This is a critical factor in the rapid advance of a technology which has brought so many new products in the relatively short time period of a hundred years. In many cases the recording engineers in film studios and their counterparts involved in producing popular songs were all working for the same business organization. The integrated, multinational corporation with interests in many different areas was the ideal vehicle to develop new technology.

The history of recorded sound provides us with an ideal case study of the causes and consequences of technological change. One thing that the tumultuous story of the phonograph tells us about technological change is that it is rarely absolute and final. Threatened men live long, and so-called obsolete technologies have managed to survive in the face of clearly superior competitors for long periods. For example, years after the microgroove record replaced the shellac 78-rpm disc, companies continued to press them and make the little steel styli which were needed to play them. And over 10 years after the CD appeared, the microgroove disc showed no sign of disappearing. This endangered species of vinyl continued to be pressed, purchased, and cherished in the 1990s. It has a small but vociferous band of supporters who refuse to accept that digital technology provides a better sound:

CDs are to records what videos are to movies: sampled, scanned, and coarse, missing huge chunks of information. . . . If you want fake, processed, artificial, lifeless, dimensionless sound . . . go spend \$15 for the privilege and buy CDs.⁶

The history of recorded sound technology is not therefore a catalogue of one smashing innovation following another, a procession of

new and improved machines with each one superior to the one it replaced. If there is one lesson a businessperson might learn from this history, it is that technology does not count for everything in this high-tech industry. Emphasizing the machine and putting technical considerations above artistic ones was the mistake made by Thomas Edison, and it proved fatal for his phonograph. Time and time again it was the mass appeal of the music recorded and not performance of the record or player that was critical in achieving commercial success.

Buying a stereo system is often an ordeal by high technology, a baffling initiation into a highly complex area of mysterious machines and puzzling acronyms: db, THD, S/N, kHz, and so on. No matter how many technical considerations are included in the analysis of sound reproduction, the enjoyment of recorded music is purely a subjective affair – there is no real way to prove that one machine or record sounds superior, other than listening to it, and everybody hears the music differently. Whatever the technical advances incorporated into the stereo, appreciating it is still relative.

There are still many people enjoying cylinder phonographs today. Devotees of acoustic recording can be found all over the world, playing their 2- and 4-minute records on well-oiled phonographs. Allen Koenigsberg's *Antique Phonograph Monthly* has a subscription list of over 2,000. Most enthusiasts use their machines to play back the sounds of the past, but several of them still use theirs to record music. The first recording studio established by Edison in his West Orange laboratory has been revived to record with acoustic phonographs. In a fine country house just outside of London, Brian Thorne and Dominic Combe use the same type of machines, and the same kind of techniques, in an acoustic recording studio. They would not claim that their recordings are superior to the compact discs made by Denon or Deutsche Grammophon, only that they get more enjoyment listening to the old cylinders. And surely that is the point.

Technology and history

The novelty of the phonograph was that it could preserve a part of life that had previously been a fleeting experience. Live music lasts only as long as the performance, for that short time when one hears the music, and after that it is just a memory. With proper care, a recording can last a lifetime or even longer, preserving the perform-



**Jenny Lind is only a memory,
but the voice of Melba can never die**

Two voices of finest, purest gold.
One is gone forever.
The other lives for all time.
There was no Victrola to capture the fleeting beauties of Jenny Lind.
But Melba's voice will still be heard in centuries to come.
To-day Melba herself thrills and entrances vast audiences throughout the world. Happy singer and happy public, that her flawless, limpid notes will flow forever in undiminished beauty from her Victor Records!

Practically every great singer and instrumentalist of this generation makes records only for the Victor—thus perpetuating their art for all time.

Victors and Victrolas, \$10 to \$400. Victor dealers everywhere.

These Victor Records demonstrated in all stores on the 1st of each month.

"Victrolas" is the Registered Trade Mark of the Victor Talking Machine Company, and the name of the record.

Important Notice: Victor Records and Victor Machines are constantly being improved by our research department. It is therefore recommended that you keep your records in a safe place, and not use them with the latest, or otherwise, records.

Victor Supremacy

Victor THE MASTER'S VOICE

Figure 0.1. Victor stresses the permanence of its recordings. (Courtesy of General Electric)

ance for many years after the musician is dead. The noun *record* is therefore an appropriate term for the product of the recording industry. It is not just a consumer product but an artifact of a time gone by.

The sound recording has brought us more than the enjoyment of music, for it is as good a piece of evidence about the past as a primary source document from an archive or a faded old photograph. Amer-

icans living in the last decades of the nineteenth century found themselves in the unique position of being able to save small slices of their existence, to capture their images on photographic film and their voices on wax cylinders. Edison and the other inventors involved in making talking machines imagined that they would be able to create libraries of sounds, not just great pieces of classical music but also messages from the important men of the day and records of the great speeches of “our Washingtons, our Lincolns, our Gladstones” which would help in the preservation of language and national values.⁷ They recorded poets and politicians as well as opera singers and pianists.

A sound recording is a piece of historical evidence. It has an impact that goes well beyond the written word or photographic image. Records enable us to listen in on history, to hear it again as it happened. The crash of the airship *Hindenburg* in New Jersey in 1937 is well preserved on film, but it is the spoken account of the disaster which brings the event to life and stirs the emotions. While only Lord Carnarvon was lucky enough to accompany Howard Carter when he entered the tomb of Tutankhamen, we can still hear Carter’s dramatic account of the great moment against the swish and crackles of a worn 78-rpm disc. The recent discovery of some phonograph cylinders reputed to be of the poet Walt Whitman caused great excitement, for it was suddenly possible to hear the poems as the poet intended them.

Businesspeople have been using dictating machines to preserve their correspondence since the 1880s. Presidents as far back as Franklin Roosevelt placed recording machines in the White House to provide an exact record of historic decision making. In 1971 President Richard Nixon made a fateful decision to install tape recorders in the White House. The Secret Service planted voice-activated machines in the Oval Office, Lincoln Sitting Room, Cabinet Office, Nixon’s private suite in the Executive Office Building, and the Presidential retreat at Camp David.

When the presence of the Nixon tapes was first made known in July 1973, at the height of the Watergate scandal, 4,000 hours of conversations had been recorded on 950 reels of tape. These tapes did more than provide undeniable evidence that the President had ordered the cover-up of the Watergate break-in; they gave a detailed and intimate picture of the Presidency with the stark honesty that other records had not been able to reveal. Although many of the expletives were deleted when transcripts of the tapes were released to

the public, they did as much to damage Nixon's reputation as the recorded conversation with H. R. Haldeman on June 23, 1972, in which they decided to deceive the FBI about the Watergate affair.

Other pieces of recorded sound which bring us into close contact with the past are the speech of President Roosevelt delivering the war resolution in Congress after Pearl Harbor and the broadcasts of Edward R. Murrow from London during World War II, but none were as important in American history as the Nixon tapes, which cost him the presidency and are still at the center of a legal wrangle.⁸

Unfortunately for historians, very few of the sounds of American history have been recorded. We have only brief aural glimpses of the past. We can go back to the early twentieth century for the first politicians' voices on record, beginning with Taft and Roosevelt (the recordings of Gladstone, Lincoln, and other nineteenth-century leaders were faked for the phonograph), but what we hear are prepared speeches and not the dialogue of decision making. We have radio broadcasts from the 1920s onwards, but very few historic events. We have the sound of Lindbergh's plane taking off from Long Island as he headed across the Atlantic in 1927 and his interviews when he returned, but nothing in between. We do not have one continuous archive of recorded sound for any aspect of American life from 1877 to the present, except that of the music and humor which was recorded for entertainment purposes.

Technology and culture

When the talking machine companies were trying to create a market for their products in the early years of the twentieth century, one of the attractions of sound recording was to bring back memories of times gone by. In advertisements they reminded the public that nothing re-creates the past like the sound of a favorite song. There is a great deal of truth in this claim, for when we listen to the music of the 1960s or the 1920s those familiar tunes help us conjure up mental images of the past.

The vast number of recordings made of popular entertainment, music, radio broadcasts, and film sound tracks from the turn of the century onwards is an important archive of popular culture. Recorded sound was a technology for the masses, and these recordings were at the heart of a new mass culture of entertainment. The content of these

records reflected the times in which they were produced, providing us with insights into those who listened and those who produced them.

Technology did more than preserve a record of a shared culture; it also influenced it. Recorded sound was the great educator, attracting generations of performers into musical careers and schooling them in styles of music which were often not written down. The recording proved to be the means of diffusing styles and bringing ethnic or regional music to a larger audience. The music it saved could go on to inspire countless young musicians. The music which went unrecorded faded from history without a trace.

Recorded sound played the critical part in the development of two uniquely American styles of popular music: jazz and rock'n'roll. The blues and jazz records made in the 1920s and 1930s formed a cultural archive of recorded sound that was a treasure trove for young musicians in the years after World War II, and not just those in the United States but the youth of the industrialized world.

The talking-machine industry was an international enterprise, with the free movement of ideas and machines across borders. The United States was always the base for the multinational companies in the nineteenth and early twentieth centuries because the advances in technology were normally made in American laboratories. Although Europe provided much of the music, America was the source of the machines. Around World War I, this international division of labor was altered; the United States became the source of the new sounds and images that transformed popular culture in the 1920s. Recorded sound was one of the foundations of this movement, which spread over the western world but retained a distinctly American character. The influential music and movies were made in the United States and distributed about the globe by American multinational companies.

The technology of recorded sound has played a very large part in the diffusion of American culture overseas. It figures prominently in what some scholars have called *cultural imperialism*. In popular music, in the sound tracks of motion pictures, and in the dialogue of television programs and advertisements, recorded sound has become the voice of the United States to the world. And nothing brings home the immediacy of the cultural experience like its sounds, whether it is the clipped sentences of Jimmy Cagney on screen or the blues shouting of Big Bill Broonzy on record. The industry that mass produces these sounds has effectively marketed its products all over the world, mak-

ing the music and speech of North Americans a global product and facilitating the diffusion of American culture. What was heard by the many millions who have listened to the scratchy 78s of Louis Armstrong, or the fading sound track of a Warner Bros. gangsters movie, was America on Record.

PART ONE
THE ACOUSTIC ERA

1. The inventors

In the last decades of the nineteenth century, the United States was at a crossroads: a sparsely populated agricultural country, only a few hundred years away from the wilderness discovered by the colonists, was about to enter a period of unprecedented economic growth to become the world's greatest industrial nation. A large part in this economic transformation was played by inventors, a group of men who devised the new technology which was the key to rapid industrialization. Their achievements were acknowledged worldwide as the "Yankee ingenuity" that transformed the United States into an industrial giant.

Of all the inventors of the nineteenth century, Thomas Edison is the best known. Statistically he holds the most U.S. patents (a record which will probably never be beaten), and mythologically he is the world's greatest inventor – the "Wizard of Menlo Park" whose magic helped create the modern industrial society in which we live. Edison became a legend in his own time; unlike most of his fellow inventors, he died rich and famous.

The invention which gained Edison his fame was the phonograph. Although nowadays we think of this machine as a mechanical entertainer, it was in fact part of a communications revolution, a revolution which brought so many dramatic changes in American life that it far exceeds any of the so-called communications or computer revolutions of the twentieth century. By 1850 inventors had developed a system of communications in which an electric current could be used between two places to replace the letter delivered by hand or by horseback. Messages now moved at the unimaginable speed of electricity. The rapid communications afforded by the telegraph became one of the foundations of American industrial growth. By the 1860s it had spawned a great international business and a host of research laboratories: the telegraph was the high-tech field of the 1860s and 1870s.¹

Edison was part of this dynamic new field of telecommunications. He was one of many ambitious young men who saw the telegraph as the most profitable field for new inventions. He worked as an itinerant telegrapher in the 1860s, moving from place to place and doing some inventing in his spare time to improve the equipment with which he worked. Another amateur inventor was Alexander Graham Bell, a teacher of the deaf in Boston, who hoped to make a better living by devising new telegraph transmitters and receivers.

The inventors came from all backgrounds and from all walks of life. Some had some experience in the mechanical arts and others (like Bell) had little but ideas and ambition. Many were recent immigrants from Europe, such as Emile Berliner, who left Germany and the threat of conscription into the Prussian army to seek his fortune in the United States in the 1870s. After a succession of jobs – working as a clerk in a store, selling glue, and painting backgrounds for photographs – he obtained employment in a chemical laboratory and started his study of physics and electricity.

Some inventors worked in laboratories of the large telegraph companies such as Western Union; others made do with makeshift apparatus in their lodgings. Those who could not afford their own work areas rented out space in machine shops, where precision tools and skilled workmen could be hired to make experimental equipment.

Invention in the late nineteenth century was not purely an intellectual affair. Men like Edison and Bell invented by doing, by turning their ideas into working models and then trying to make them work. The critical flash of insight often came in operating their apparatus.² Experimental transmitters and receivers had to be made up by skilled craftsmen before any new idea could be tested. Edison frequented the shop of Charles Williams in Boston, where he went to experiment after finishing work at the Western Union office. A few years later Bell had his devices made at this same shop, where he made the acquaintance of a mechanic named Thomas Watson, who was to figure large in his future.³

The appeal of the telegraph was twofold for the aspiring inventor. First, an improved transmitter or receiver was worth a lot of money, as was evident in the lucrative prices Edison obtained for his quadruplex telegraph patents. He used the income from various research contracts to build a laboratory in Menlo Park, New Jersey, and this set him along a path that was to lead to some of the most astonishing

inventions of the nineteenth century. Edison's rise from penniless tinkerer to well-known inventor with his own laboratory was an inspiration to others who hoped to emulate his success.

The other attraction of the telegraph was that this technology was in its infancy and there were more valuable inventions to be made. The messages could not be sent very quickly or very far, and only one at a time went along the wire. Edison's quadruplex idea enabled four messages to be sent on one line and therefore increased the capacity of the network four times. He also worked on an automatic telegraph – a device that stored telegraph code on punched tape, which could then be run through the transmitter at high speeds – to quicken the pace of communication.

These innovations, and many more like them made by other inventors, brought about a more efficient use of the capital equipment of the telegraph companies and improved the service they offered to the customer. But they did not alter the fundamental basis of telegraphic communication; it was still dots and dashes created by interrupting the flow of current from transmitter to receiver. Some visionary inventors devoted their efforts to the speaking telegraph, which could transmit and receive the sound of the human voice – a task so difficult that many considered it to be impossible. Charles Bourseul of France had tried to transmit speech over wires around 1854 but had failed, as did many other inventors. The quantum leap to use electric currents to transmit the sound of the human voice was first accomplished by accident, and by a rank amateur in the business of invention.

The secret of electric speech

During the 1870s Alexander Graham Bell worked on several ideas to improve the telegraph. He was interested in the harmonic telegraph, which could send more than one message along the wire. It used tuning forks or steel reeds that were resonant at different frequencies to transmit dots and dashes as signals of these different frequencies along the same wire. At the end of the wire were several receivers, each tuned to a different frequency; each one responded selectively to the incoming messages.

Bell was experimenting with one of these devices in 1875. It consisted of a group of steel reeds each attached to an electromagnet, which acted as a transmitter by making and breaking a circuit. At the

other end of the wire was a corresponding receiver arrangement of reeds and magnets. The steel reed was a thin slice of metal attached to one pole of the electromagnet with its other end just projecting over the opposite pole of the magnet. While Bell was experimenting with magnetizing the reed and creating varying currents, he accidentally discovered that plucking the reeds next to the electromagnet would produce a current which would activate the magnet at the other end, causing the reeds attached to it to vibrate and produce an audible sound.

Bell had three of these transmitters in one room of the attic of Williams' shop, with three receivers in another room attended by his assistant, Watson. As Bell activated the current in each transmitter in turn, the corresponding reed in the receiver in the next room sounded, except for the last one. The reed had probably become stuck to the pole of the electromagnet. With the current turned off, Bell shouted to Watson to free the reed, and as Watson plucked the reeds Bell saw that his own reed was vibrating. He excitedly told Watson to keep plucking the reeds and then, moving close to his transmitter, he listened to the sound of them vibrating, convinced that what he heard was the pitch and the amplitude of that sound in Watson's room. His biographer, Catherine Mackenzie, described this moment of insight: "And then, . . . by accident, and for a fraction of a second, the clue to electric speech flashed into the noise and heat of that dusty Boston attic."⁴

Although not as dramatic as Mackenzie described it, this event did persuade Bell that it might be possible to transmit the sound of speech along a wire. He told Watson that if he could find a way of making a current of electricity vary in its intensity as air varies in density as sound passes through it, "I can telegraph any sound, even the sound of speech."⁵ He experimented with the transfer of the mechanical movement of sound in the air into the variation of a minute electric current produced by electromagnetism. This research eventually led him to the conclusion that the complexities of pitch and timbre in human speech could be transmitted electrically – the waves caused by sound were analogous to the variable waves of electricity. He hoped that a current of electricity could be produced that varied in intensity precisely and exactly as the sound waves varied.

Once he grasped the principle of transforming sound waves into electric currents, Bell asked Watson to build an instrument that used

a thin membrane diaphragm with a metallic center instead of the reed. The pressure of the sound hitting the diaphragm was faithfully duplicated by the moving sliver of metal as it interacted with a magnetic field to produce a varying current. In reverse, an electromagnet next to the diaphragm could vibrate it – reproducing sounds by its subtle movement – as current passed through its windings. Throughout 1875 Bell and Watson worked to perfect the telephone transmitter before applying for a patent. By February 1876 they had a model in which the electric current was provided by a battery; this current was modulated by the movement of the diaphragm.

On the evening of March 10, 1876, Bell and Watson were experimenting with this device in their boarding house: Bell at the transmitter in one room and Watson with the receiver in another. This was the night when Watson heard the famous words “Watson, come here, I want you!” – the first words to be transmitted on the telephone.⁶ Throughout the summer of 1876, the two men developed their ideas. They built a new form of telephone transmitter which used a diaphragm attached to the poles of an electromagnet by a metal armature. This magneto-telephone was publicly demonstrated to several groups in 1876. Bell took out a patent for his invention and began to talk with businessmen about constructing a communications system that would compete with the wires of Western Union.

Bell's experiments in Boston followed the work of many others in the realm of electricity and acoustics. His telephone was based on the scientific principles of electromagnetism developed by Hans Oersted in Denmark in 1820, and on Michael Faraday's study of the induction of electric current, which was published in England in 1831. Bell's research was not much different from that carried out by aspiring inventors all over Europe and the United States, but Bell came to the important conclusion first. As his biographer, Robert Bruce, has pointed out: “In retrospect, it seems obvious that if the magnet were brought close to a flexible metal diaphragm and a variable current passed through the coil (of the electromagnet), the magnet would cause the diaphragm to vibrate to produce sound.”⁷

Bell was not alone in his experiments on the harmonic telegraph. Thomas Edison also saw great potential in this technology and took out several patents on it. In Chicago, Elisha Gray worked on harmonic (or acoustic) telegraphy as a means to send a number of messages simultaneously. An ex-carpenter with two years of college to

his credit, Gray had become a professional inventor and manufacturer of telegraph instruments. Gray succeeded in transmitting sounds by the same methods that Bell used to transmit speech. By the time that Bell moved to Boston, the term *telephone* had already been applied to an acoustic telegraph invented by Philip Reis of Germany in 1861. This device used the vibrations of sound to move a diaphragm that activated a make-and-break telegraph circuit. Although Reis believed that his device could transmit the sound of the human voice, like Gray he concentrated on sending musical tones.

Bell managed to beat his many competitors in this field by being the first to apply for a patent on February 14, 1876, just a few hours before Elisha Gray arrived at the patent office to file a preliminary application for virtually the same idea. But Bell was first, and his triumph was complete.

The first public demonstrations of Bell's telephone were accompanied by incredulity and amazement. The squeaky voice emerging from the wooden box that held the assembly of wires, the metal diaphragm, and electromagnet was a bewildering herald of a new era in communications. Instead of by dots and dashes, messages could now be sent exactly the way they were spoken. The revelation that sound could be transmitted along a wire caused consternation in the board rooms of the telegraph industry and unbridled excitement among the international fraternity of inventors.

The widely dispersed group of experimenters who had been working on harmonic and speaking telegraphs were electrified by the news of Bell's telephone. The pioneering experiments of Bell, Gray, and Reis were quickly replicated in hundreds of workshops and laboratories. Bell's apparatus was far from being a commercial technology; the sound from the telephone receiver was faint, and it worked only over short distances. There was plenty of room for improvement. At the Menlo Park laboratory, Edison devised a better telephone transmitter, which sent a clearer and stronger sound to the receiver. Berliner followed the same path in his workshop in Washington.

In the years that followed Bell's invention of the telephone, there was a worldwide study of variations in electric currents that could reproduce sounds. The thin metal diaphragm and electromagnet assemblies were put together, tested, examined, and rebuilt. Many more amateur and professional inventors grew familiar with the delicate electromechanical devices that picked up sound waves and turned

them into electric currents. Increased familiarity with the diaphragms and electromagnetic assemblies, which turned the faint vibrations of sound waves into undulating currents, was to bear fruit in the flood of new telephone transmitters and receivers which were submitted to the patent office.

These experiments were also to prepare the ground for the development of a technology for recording sound, and it is certainly not a coincidence that technology to save telephone messages occurred only a year after Bell transmitted those famous words to Watson.

Saving and reproducing sound

The requirement to record and reproduce sound was a direct consequence of this revolution in communications. With millions of important messages now flowing from place to place, it became desirable to capture and store them for reference. Saving incoming messages and replaying them later at a comfortable speed also made it possible to use unskilled operators to cope with the exponential increase in messages.

The automatic telegraph achieved both of these goals. Morse code was punched into a strip of paper in the form of holes. These holes made electrical contact in the transmitter which caused breaks in the circuit. The strips of paper could be run through the transmitter at very high speeds. At the receiving end, the electrical pulses were recorded onto a strip of paper either by electrochemical changes or by an electrically activated ink recorder.

Thomas Edison had plenty of experience in automatic telegraphy when he opened his Menlo Park laboratory in 1876. He wanted this first industrial research facility to produce a stream of useful innovations, practical inventions that would meet a well-defined need in commerce or industry. Consequently, he directed his experiments into the new field of electrical communications. An improved telephone was one of his priorities because Western Union wanted their own machine to help ward off the threat from Bell's telephone.

Edison had already covered a great deal of ground in his research on the telegraph and telephone. He knew how to turn sound waves into mechanical movement that could induce electric currents. He knew how to manipulate electromagnetic attraction to move switches and make circuits. He had studied the work of many other inventors

in the field, examining obscure telegraph patents in European journals. He was well versed in the theories of the German scientist Hermann Helmholtz on the nature of sound waves. He was also aware of the work of Leon Scott in France, whose “phonautograph” used a stylus attached to a membrane diaphragm to trace out the undulations of sound waves on a cylinder covered with smoked paper. The development of the telephone transmitter familiarized Edison with the thin discs of metal or animal membrane which acted as a diaphragm by vibrating to produce sound waves. All these pieces of information were to be brought together in the invention of the phonograph.

As was usual in his laboratory, Edison directed his teams of experimenters to work on several related projects in 1877. He was engaged in the job of producing an improved telephone for Western Union. In addition to this “speaking” telegraph, Edison and his men worked on harmonic and automatic telegraphs. During the summer of 1877, he took some time off from the telephone to sketch out a few ideas for recording incoming messages. As a commercial inventor he realized the value of saving telephone messages, especially as he correctly anticipated that businessmen would be the first to use them to replace written communication.

The devices he drew were inspired by his earlier inventions of automatic telegraphs that employed strips of treated paper marked with the dots and dashes of Morse code. He made one important addition. In previous experiments he had been struck by the force of the sound waves moving a diaphragm and wondered if a needle attached to the center of a diaphragm could be powered by sound to make a mark. One night in July 1877, his staff rigged up an indenting stylus connected to a diaphragm, which in turn was attached to a telephone speaker. As Edison shouted into the speaker, a strip of paraffin-coated paper was run underneath the stylus. An examination of the strip showed the irregular marks made by the sound waves. When the strip was pulled back under the stylus, the group of men crowded around the laboratory table heard, with disbelief, the faint sounds of Edison’s shouts of a few minutes previous. His chief assistant, Charles Batchelor, remembered that the sounds were not by any means a perfect reproduction, “but the shape of it was there, and so like the speaking that we all let out a yell of satisfaction and a ‘Golly it’s there’ and all shook hands.”⁸

The next day Edison made note of the experiment and wrote that

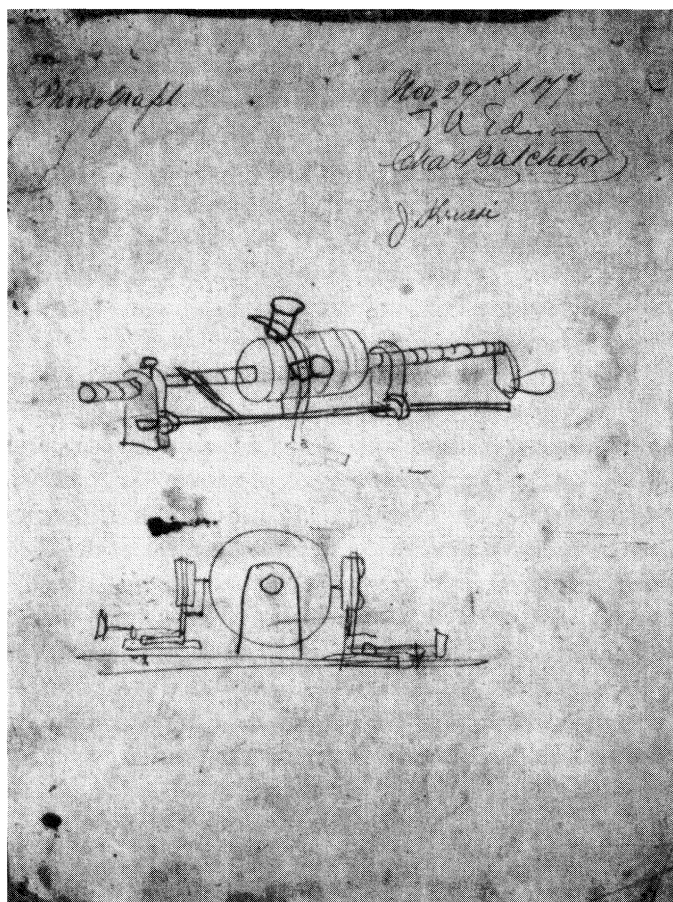


Figure 1.1. Edison's sketch of the phonograph, reputedly of November 1877. (U.S. Department of the Interior, National Park Service, Edison National Historic Site [hereafter cited as ENHS])

"the spkg vibrations are indented nicely & theres no doubt that I shall be able to store & reproduce automatically at any future time the human voice perfectly."⁹

This amazing claim was not made into reality until December 1877 when the laboratory workers made up a working phonograph. It consisted of a cylinder attached to a long feed screw turned by a hand crank. A sheet of tinfoil was carefully wrapped around the cylinder.

On one side of the cylinder was the recording assembly, a funnel-like mouthpiece (similar to the one used for the telephone) connected to a thin metal diaphragm to which a steel stylus was attached. Shouting into the funnel vibrated the diaphragm and moved the stylus, which incised the spiral analog of the sound waves onto the soft metal foil as it was turned. On the other side of the cylinder, a playback reproducer of stylus and diaphragm was run back over the indentations in the foil to vibrate the diaphragm in the same way that the recording had been made. The listener had to pay close attention to discern the faint noises coming from the vibrating diaphragm.

Demonstrated at the office of *Scientific American* on December 6, 1877, the phonograph quickly became the rage. As word got out, people flocked to the building in New York City and packed into the room to hear the machine talk. When the news appeared in the press, train loads of people, “great crowds of people” as Batchelor remembered, arrived at the laboratory to hear “with astonishment” the sounds of their own voices.¹⁰ Even the President of the United States wanted to hear the talking machine.

The acclaim given to the phonograph has tended to obscure the fact that although this was a marvelous invention, it was more of a step forward along a well-travelled path than a great breakthrough.¹¹ As with many of Edison’s inventions, part of its achievement was to bring together several ideas. It was an application of the work done by Leon Scott and others. The phonautograph made a tracing of the sound waves as they vibrated the diaphragm. Edison took this idea one step further and found that the impression of the sound could be used to reproduce the original. The inventors who had been working in the same areas as Edison had the idea at their fingertips but did not see it. As Bell wrote in 1878: “It is a most astonishing thing to me that I could possibly have let this invention slip through my fingers.”¹²

Bell must have been astonished not only because he had been working along the same lines as Edison and not seen how sound waves could be mechanically reproduced, but also because the phonograph was such a simple idea. Edison’s achievement was to establish that sounds could be recorded and re-created with just the mechanical power of the stylus vibrating as it moved up and down on the minute indentations on the paper or tinfoil. In considering the possibilities of recording sound, Edison thought that saving messages would require

the same transformation of sound waves into electrical impulses that had been successfully used in the telephone. It made sense that the method used to create electric speech might be the means to save and retrieve it.

The critical experiments carried out in Menlo Park in the summer and fall of 1877 proved that the storage and recovery of sound could be done without the electrical transformation used in the telephone. There was a simpler way, and Edison had found it. When he announced that he had devised a “wonderfully simple” apparatus for recording the human voice, for once he was not exaggerating.

Although the tinfoil phonograph made an enormous impression on the public and on the fraternity of inventors, it had very little commercial potential. Edison, the most commercially minded of all inventors, had stumbled on it in the pursuit of other goals. He called it an “invention pure and simple” that he had discovered rather than methodically sought out. He saw it in terms of a device to save incoming phone calls, and as such it was the first attempt to make a telephone answering machine.¹³ He also saw it as a means to increase the length of telephone transmission by situating it at the end of one line and then playing a recording of a message on another line. The phonograph acted as a telephone “repeater.” But since only about a minute or two of sound could be saved by the tinfoil phonograph, it could not serve practically in either function.

Several other problems had to be solved before it could be considered a commercial machine. The tinfoil could be used only two or three times before it deformed and the message was lost. The intelligibility of the replayed sound depended on the even turning of the crank, for even a slight change in speed distorted the sound beyond recognition. This was not an easy machine to operate. Ed Pershey, the curator of the Edison National Historic Site for some years, has considerable experience coaxing a recording out of the tinfoil phonograph. He commented that “Although the process is, in essence, simple, it takes dexterity and a fair amount of luck to achieve success.”¹⁴

The disembodied sounds and squeaks emerging from the tinfoil could be discerned by the listener, but it took practice to recognize speech. Edison had demonstrated a principle, but his phonograph could not save and reproduce sounds accurately or consistently enough to be useful to the casual listener. In the years after 1877, Bell

and his associates made the telephone one of the most profitable inventions of the nineteenth century, but the phonograph remained a novelty, a scientific toy.

The talking machine takes shape

Edison could choose from several formats in his development of the talking machine. In addition to the cylinder, there was the revolving disc and the strip recording. The staff of the Menlo Park laboratory tried out several types of disc (or plate) records, in which the sound waves were incised in a spiral on the flat surface of the disc. Edison had plenty of experience in using strips of paper or foil as recording material in prior experiments in telegraphy. In his patent of December 1877, he described the strip of tinfoil which passed under the stylus and received the indentations of sound waves in the form of a continuous groove.¹⁵

With three possible formats to choose from, Edison concentrated on the cylinder; it was simple to build, and he was familiar with the revolving cylinder and stylus arrangement from his work on telegraphy and from the numerous lathes that were found in his laboratory. The way that the stylus scratched the revolving cylinder was exactly the manner in which the cutting tools of lathes shaved off ribbons from revolving pieces of metal. It was therefore familiarity which determined that the cylinder model would be exhibited to the world as the phonograph.

In the winter of 1877/1878, the experimenters at Menlo Park were hard at work improving the talking machine. They even built a disc model which had better sound reproduction than the original cylinder version, and Edison wrote confidently to his financial backers that “a practical marketable machine” was within his reach.¹⁶

Yet this did not happen for many years. Edison’s attention never stayed on one subject for very long, and he was soon drawn to other experimental projects, such as improving his telephone transmitter for Western Union. In late 1878, at a critical point in recorded-sound experiments, Edison stopped work on electrical communications to concentrate on the development of a high-resistance incandescent electric light, which was an invention with a clearly defined use and a great potential market. The same could not be said for the tinfoil phonograph, and it languished in the Menlo Park laboratory while

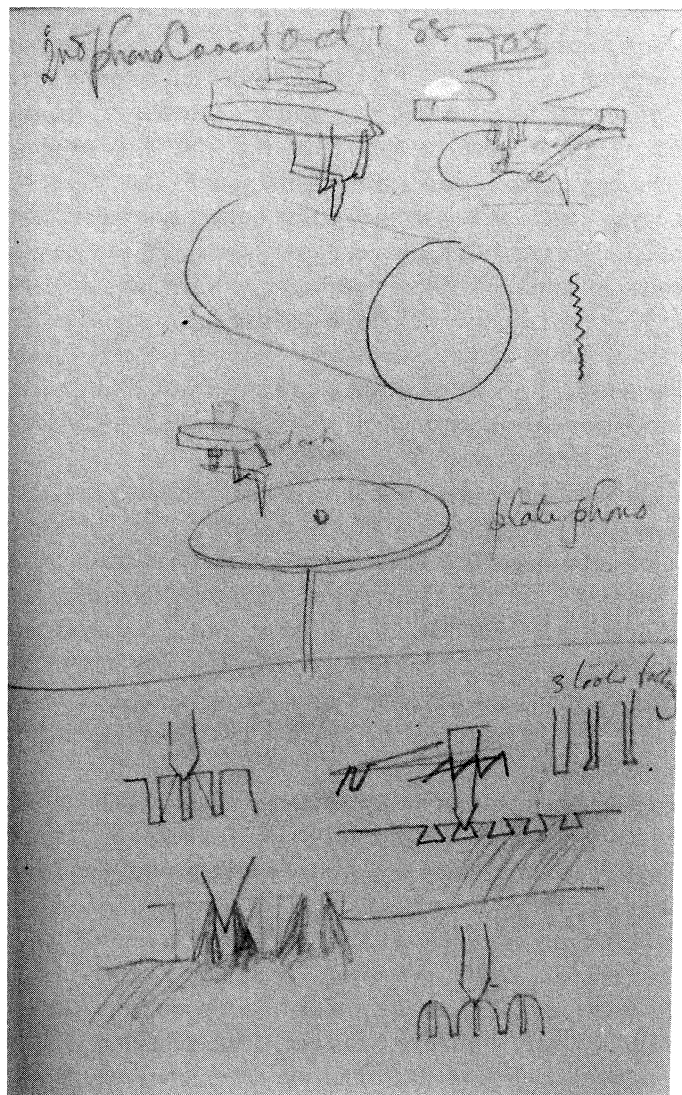


Figure 1.2. Edison's idea for a disc or "plate" phonograph in 1888.
(Author)

Edison set up the first system of electricity supply from a central power station in New York City. The task of turning his phonograph invention into a practical device therefore fell to other inventors.

Alexander Graham Bell had almost stumbled on the phonograph idea in 1874, when he speculated that the phonautograph format could be used to re-create the original sound. But Edison beat him to it, as he also beat Bell to the carbon button transmitter that dramatically improved the telephone. Nevertheless, Bell saw commercial potential in sound recording and began to think about it when he set up his own laboratory in 1880 to investigate the science of sound. He was joined in his Volta laboratory by his cousin Chichester Bell, who was well versed in physics and chemistry, and Charles Tainter, a precision instrument maker who had worked in Charles Williams' shop making telegraph equipment.

Together they experimented on acoustic and electrical technologies. They even investigated the idea of recording sound photographically. Bell demonstrated in 1880 that it was possible to transmit speech over a beam of light. The "Photophone" was later developed at the Volta laboratory. In 1884 and 1885 Bell and Tainter made experiments on recording sound as spiral waves on glass discs coated with emulsion. They manipulated the sound to cause variations in a light source which they recorded on the sensitive photographic emulsion on the revolving disc.¹⁷

The most important project undertaken at the Volta laboratory was the development of a commercial machine based on Edison's phonograph. They substituted wax for the tinfoil on the cylinder and coated a layer around a cardboard core. While the phonograph's needle displaced the recording medium, indenting the sound wave upon it, Tainter's idea was to cut or engrave an impression on it. The end of the cutter was shaped like a chisel, and it made a cleaner cut than the indenting stylus. The result was a more intelligible sound and a little less background noise. The wax cylinder was much more durable than the tinfoil and could be taken on and off the machine without deforming the groove of the sound wave.

Bell and Tainter's machine was arranged the same way as the phonograph, except that there were two parallel axes, one for the cylinder and one for the lead screw which held the reproducer. The recording cutter and reproducing arm were moved precisely along the wax cylinder by a lead screw assembly, which ran parallel to the turning axis

of the cylinder. It worked like the feedscrew of a lathe. Tainter tried to balance the reproducing stylus assembly so that it would float in the groove, enabling it to react to the tracing of the sound waves while riding out the imperfections on the wax surface.¹⁸

The improved talking machine developed in the Volta laboratory was named the *graphophone*. In terms of ease of operation and fidelity of sound reproduction, it was a vast improvement on the phonograph. After many years of experiments, the three Volta associates obtained several patents in 1886 and took their machine to Edison with a proposition to join together and commercially exploit the new technology. But Edison would have none of it, and smarting from the slight of having others improve his famous invention, he returned to his phonograph. After years of neglect he was finally going to perfect the recording of sound.¹⁹

In 1887 Edison finished the construction of a large laboratory in West Orange, New Jersey. Shortly after the facility opened at the end of the year, he demonstrated an improved phonograph. The wax cylinder and general arrangement of this model owed a lot to the graphophone, but instead of a hand crank it was powered by an electric motor which was placed underneath the revolving cylinder. It drove the cylinder spindle through a system of bevelled wheels. As the cylinder turned, a lead screw moved the reproducer along the cylinder's groove.

The recording and reproducing diaphragms were joined together in the form of a pair of spectacles. The operator shouted into a small horn attached to the recorder, and the vibrations of the diaphragm moved the sharp tip of the recording stylus. After it had cut the groove of the sound wave into the soft wax cylinder, the spectacle frame was then turned to bring the reproducer stylus onto the groove of the cylinder, and the user listened to the recording through an eartube connected to the reproducer.²⁰

Edison and his experimental teams at the West Orange laboratory continued to experiment on sound recording and to improve the talking machine. After weeks of intensive work in the summer of 1888, Edison announced that he had finally perfected the phonograph. It now came in a rectangular box which enclosed the electric motor. The cylinder and reproducer assemblies were fixed to a metal plate on top of the box. The most prominent feature of the machine was the tapered mandrel – a shaft of metal which held the cylinder – which

ran along the top plate. After sliding a cylinder on the mandrel, the user swung the diaphragm assembly onto it and started it turning.²¹

During this time Edison made frequent pronouncements from the West Orange laboratory that the talking machine was now ready for commercial introduction. But these were no more than a smoke screen to give him time to recover lost ground in the technology of recorded sound. Many of the international fraternity of inventors, who had once applied their talents to the telephone, were now fully engaged in recording sound. Edison's English patents on the phonograph had lapsed in 1885, leaving the field wide open on both sides of the Atlantic to the many newcomers who thought they could produce a better machine. Most of their efforts were variations of the basic phonograph configuration, but one design broke away from the cylinder idea.

The gramophone used a disc instead of a cylinder. It was the work of Emile Berliner, who demonstrated it in 1888. It was powered by a hand crank and had the same kind of diaphragm and stylus assembly that could be found on any talking machine. The gramophone did not employ a lead screw to move the stylus along the groove in playback; instead the spiral groove in the disc gently pushed the stylus across the face of the record. Where Edison and Tainter had employed a vertical (up and down) cut in recording sound waves onto the cylinder, Berliner employed a lateral cut in which the stylus moved from side to side.

In 1888 and 1889 the stream of talking-machine patents produced in the United States and Europe turned into a flood. At the West Orange laboratory, Edison led his teams of researchers in an intense campaign to turn a laboratory device into a dictating machine that could be used by businessmen. Bell and Tainter were doing the same thing in Washington. They faced a multitude of problems. The hand crank that turned the cylinder or disc had to be turned at constant speed or the pitch of the reproduction changed noticeably. This was asking too much of the average user and consequently another power source had to be found. The inventors experimented with electric motors, clockwork and spring motors, foot treadles, gravity feeds (in which a weight pulled down a cord attached to a spindle), and even water power.

The other major concern was the reproduction of sound. There were endless possible variations in the critical assembly of stylus and